

HEFT Feynman rules

85 vertices, generated by FeynRules from LHEFT.

Physical fields

Symbol	L ^A T _E X	Description
A	A	photon
b	b	bottom quark
bbar	$\bar{b}$	bottom antiquark
c	c	charm quark
cbar	$\bar{c}$	charm antiquark
d	d	down quark
dbar	$\bar{d}$	down antiquark
e	e	electron
ebar	$\bar{e}$	positron
G	g	gluon
H	H	Higgs
mu	μ	muon
mubar	$\bar{\mu}$	antimuon
s	s	strange quark
sbar	$\bar{s}$	strange antiquark
t	t	top quark
ta	τ	tau
tabar	$\bar{\tau}$	antitau
tbar	$\bar{t}$	top antiquark
u	u	up quark
ubar	$\bar{u}$	up antiquark
ve	ν_e	electron neutrino
vebar	$\bar{\nu}_e$	electron antineutrino
vm	ν_μ	muon neutrino
vmbar	$\bar{\nu}_\mu$	muon antineutrino
vt	ν_τ	tau neutrino
vtbar	$\bar{\nu}_\tau$	tau antineutrino
W	W^+	W boson
Wbar	W^-	W boson (conjugate)
Z	Z	Z boson

Indices

Type	Symbol	Description
Colour	m	SU(3) colour-triplet (quark) index, 1..3
Generation	f	Fermion generation index, 1..3
Gluon	a	SU(3) colour-octet (gluon) index, 1..8
Lorentz	μ	Lorentz vector index
Spin	s	Dirac spin index, 1..2

Vertices

3-point vertices

- H_1, H_2, H_3

$$-3 i \sqrt[4]{2} \text{Kappa3H} \text{MH}^2 \sqrt{G_f}$$

- g_1, g_2, H_3

$$i \sqrt[4]{2} \text{CGHNLO} \text{n1 ChiralOrder} \text{p}_1 \cdot \text{p}_2 \sqrt{G_f} \delta_{a_1, a_2} \eta_{\mu_1, \mu_2} -$$

$$i \sqrt[4]{2} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \delta_{a_1, a_2} \text{p}_1^{\mu_2} \text{p}_2^{\mu_1}$$

- g_1, g_2, g_3

$$\frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_1^{\alpha 1} \epsilon_{\mu_1, \mu_2, \mu_3, \alpha 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_2^{\alpha 1} \epsilon_{\mu_1, \mu_2, \mu_3, \alpha 1}}{\sqrt{\pi}} +$$

$$\frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_3^{\alpha 1} \epsilon_{\mu_1, \mu_2, \mu_3, \alpha 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_1^{\beta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \beta 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_2^{\beta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \beta 1}}{\sqrt{\pi}} +$$

$$\frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_3^{\beta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \beta 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_1^{\gamma 1} \epsilon_{\mu_1, \mu_2, \mu_3, \gamma 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_2^{\gamma 1} \epsilon_{\mu_1, \mu_2, \mu_3, \gamma 1}}{\sqrt{\pi}} +$$

$$\frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_3^{\gamma 1} \epsilon_{\mu_1, \mu_2, \mu_3, \gamma 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_1^{\delta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \delta 1}}{\sqrt{\pi}} + \frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_2^{\delta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \delta 1}}{\sqrt{\pi}} +$$

$$\frac{\alpha_s^{3/2} \theta_s f_{a_1, a_2, a_3} \text{p}_3^{\delta 1} \epsilon_{\mu_1, \mu_2, \mu_3, \delta 1}}{\sqrt{\pi}} + 2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} \text{p}_1^{\mu_3} - 2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} \text{p}_2^{\mu_3} -$$

$$2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} \text{p}_1^{\mu_2} + 2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} \text{p}_3^{\mu_2} + 2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} \text{p}_2^{\mu_1} -$$

$$2 \sqrt{\pi} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} \text{p}_3^{\mu_1}$$

- $\bar{b}_1, b_2, H_3$

$$- \frac{i \text{KappaQB} \text{MQB} \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$

- $\bar{d}_1, d_2, H_3$

$$- \frac{i \text{KappaQD} \text{MQD} \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$

- $\bar{s}_1, s_2, H_3$

$$- \frac{i \text{KappaQS} \text{MQS} \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$

- $\bar{e}_1, e_2, H_3$

$$- \frac{i \text{KappaLE} \text{MLE} \delta_{s_1, s_2}}{\text{vev}}$$

- $\bar{\mu}_1, \mu_2, H_3$

$$- \frac{i \text{KappaLM} \text{MLM} \delta_{s_1, s_2}}{\text{vev}}$$

- $\bar{\tau}_1, \tau_2, H_3$

- $$-\frac{i \text{KappaLT MLT } \delta_{s_1, s_2}}{\text{vev}}$$
- $\bar{c}_1, c_2, H_3$
- $$-\frac{i \text{KappaQC MQC } \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$
- $\bar{t}_1, t_2, H_3$
- $$-\frac{i \text{KappaQT MQT } \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$
- $\bar{u}_1, u_2, H_3$
- $$-\frac{i \text{KappaQU MQU } \delta_{m_1, m_2} \delta_{s_1, s_2}}{\text{vev}}$$
- A_1, W^+_2, W^-_3

$$\begin{aligned}
& -\frac{i \sqrt[4]{2} \eta_{\mu_1, \mu_2} \sqrt{\text{MZ}^2 G_f} p_1^{\mu_3} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)}}{\text{MZ}^2 G_f} \\
& i \sqrt[4]{2} \eta_{\mu_1, \mu_2} \sqrt{\text{MZ}^2 G_f} p_2^{\mu_3} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)} + \\
& \frac{\text{MZ}^2 G_f}{\text{MZ}^2 G_f} \\
& i \sqrt[4]{2} \eta_{\mu_1, \mu_3} \sqrt{\text{MZ}^2 G_f} p_1^{\mu_2} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)} - \\
& \frac{\text{MZ}^2 G_f}{\text{MZ}^2 G_f} \\
& i \sqrt[4]{2} \eta_{\mu_1, \mu_3} \sqrt{\text{MZ}^2 G_f} p_3^{\mu_2} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)} - \\
& \frac{\text{MZ}^2 G_f}{\text{MZ}^2 G_f} \\
& i \sqrt[4]{2} \eta_{\mu_2, \mu_3} \sqrt{\text{MZ}^2 G_f} p_2^{\mu_1} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)} + \\
& \frac{\text{MZ}^2 G_f}{\text{MZ}^2 G_f} \\
& i \sqrt[4]{2} \eta_{\mu_2, \mu_3} \sqrt{\text{MZ}^2 G_f} p_3^{\mu_1} \sqrt{-\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) - \text{MZ} G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}) + \text{MZ} G_f \right)} - \\
& \frac{\text{MZ}^2 G_f}{\text{MZ}^2 G_f}
\end{aligned}$$

- H_1, W^+_2, W^-_3

$$\begin{aligned}
& \frac{4 i 2^{3/4} \pi \text{aEWM CF Tn0 ChiralOrder KappaV MZ}^2 \sqrt{G_f} \eta_{\mu_2, \mu_3}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM CF Tn0 ChiralOrder KappaV MZ} \eta_{\mu_2, \mu_3} \sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}{\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \\
& \frac{2 i \sqrt[4]{2} \text{CF Tn0 ChiralOrder KappaV MZ}^4 G_f^{3/2} \eta_{\mu_2, \mu_3}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\
& \frac{2 i \sqrt[4]{2} \text{CF Tn0 ChiralOrder KappaV MZ}^3 \sqrt{G_f} \eta_{\mu_2, \mu_3} \sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM KappaV MZ}^2 \sqrt{G_f} \eta_{\mu_2, \mu_3}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \frac{2 i 2^{3/4} \pi \text{aEWM KappaV MZ} \eta_{\mu_2, \mu_3} \sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}{\sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \\
& \frac{i \sqrt[4]{2} \text{KappaV MZ}^4 G_f^{3/2} \eta_{\mu_2, \mu_3}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{i \sqrt[4]{2} \text{KappaV MZ}^3 \sqrt{G_f} \eta_{\mu_2, \mu_3} \sqrt{G_f} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}}
\end{aligned}$$

- W^+_1, W^-_2, Z_3

[illegible]

$$\begin{aligned}
& \frac{2 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder } \sqrt{G_f \text{MZ}^2} \pi p_3^{\mu_1} \eta_{\mu_2, \mu_3}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_1^{\mu_3} \eta_{\mu_1, \mu_2}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}} - \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_2^{\mu_3} \eta_{\mu_1, \mu_2}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}} - \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_1^{\mu_2} \eta_{\mu_1, \mu_3}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}} + \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_3^{\mu_2} \eta_{\mu_1, \mu_3}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}} + \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_2^{\mu_1} \eta_{\mu_2, \mu_3}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}} - \\
& \frac{2 i 2^{3/4} \text{aEWM } \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} p_3^{\mu_1} \eta_{\mu_2, \mu_3}}{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi) \text{MZ}}
\end{aligned}$$

- H_1, Z_2, Z_3

$$\begin{aligned}
& -4 i \sqrt[4]{2} \text{CFTn0 ChiralOrder KappaV } \text{MZ}^2 \sqrt{G_f} \eta_{\mu_2, \mu_3} + \\
& 2 i \sqrt[4]{2} \text{CFTn1 ChiralOrder } \text{MZ}^2 \sqrt{G_f} \eta_{\mu_2, \mu_3} + 2 i \sqrt[4]{2} \text{KappaV } \text{MZ}^2 \sqrt{G_f} \eta_{\mu_2, \mu_3}
\end{aligned}$$

- $\bar{c}_1, c_2, A_3$

$$\frac{2 i \sqrt[4]{2} \delta_{m_1, m_2} \sqrt{\text{MZ}^2 G_f} \sqrt{-\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \text{MZ } G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ } G_f \right)} \gamma_s}{3 \text{MZ}^2 G_f}$$

- $\bar{t}_1, t_2, A_3$

$$\frac{2 i \sqrt[4]{2} \delta_{m_1, m_2} \sqrt{\text{MZ}^2 G_f} \sqrt{-\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \text{MZ } G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ } G_f \right)} \gamma_s}{3 \text{MZ}^2 G_f}$$

- $\bar{u}_1, u_2, A_3$

$$\frac{2 i \sqrt[4]{2} \delta_{m_1, m_2} \sqrt{\text{MZ}^2 G_f} \sqrt{-\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \text{MZ } G_f \right)} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ } G_f \right)} \gamma_s}{3 \text{MZ}^2 G_f}$$

- $\bar{c}_1, c_2, Z_3$

$$\begin{aligned}
& - \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu_3} \cdot P_{-s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} + \\
& \frac{i \sqrt[4]{2} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu_3} \cdot P_{-s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} + \frac{2 i \sqrt[4]{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu_3} \cdot P_{+s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} - \\
& \frac{2 i \sqrt[4]{2} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu_3} \cdot P_{+s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM } \pi)} -
\end{aligned}$$

$$\frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi) \text{MZ}}$$

- $\bar{u}_1, u_2, Z_3$

$$\begin{aligned} & - \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \\ & \frac{i \sqrt[4]{2} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \frac{2 i \sqrt[4]{2} \text{CFTn0 ChiralOrder} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{2 i \sqrt[4]{2} G_f \sqrt{G_f \text{MZ}^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} \text{MZ}^2}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{2 i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{G_f \text{MZ}^2} \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} \text{MZ}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \\ & \frac{2 i \sqrt[4]{2} \sqrt{G_f \text{MZ}^2} \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} \text{MZ}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{2 i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{G_f \text{MZ}^2} \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} \text{MZ}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \\ & \frac{2 i \sqrt[4]{2} \sqrt{G_f \text{MZ}^2} \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} \text{MZ}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f \text{MZ}^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \\ & \frac{2 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f \text{MZ}^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} + \\ & \frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f \text{MZ}^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{4 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f \text{MZ}^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - \\ & \frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi) \text{MZ}} - \\ & \frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f \text{MZ}^2} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi) \text{MZ}} \end{aligned}$$

- $\bar{c}_1, b_2, W^+_3$

$$\begin{aligned} & - \frac{i \text{CFTn0 ChiralOrder} \text{MZ}^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{2,3} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ} G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \\ & \frac{i \text{CFTn0 ChiralOrder} \text{MZ} \delta_{m_1, m_2} V^{\text{CKM}}_{2,3} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ} G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \\ & \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder} \delta_{m_1, m_2} V^{\text{CKM}}_{2,3} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ} G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ & \frac{i \sqrt[4]{2} \text{MZ}^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{2,3} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ} G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ & \frac{2 i 2^{3/4} \pi \text{aEWM} \delta_{m_1, m_2} V^{\text{CKM}}_{2,3} \sqrt{\text{MZ} \left(\sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + \text{MZ} G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} \end{aligned}$$

- $\bar{t}_1, d_2, W^+_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{3,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} V^{\text{CKM}}_{3,1} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} V^{\text{CKM}}_{3,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{3,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} V^{\text{CKM}}_{3,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{t}_1, s_2, W^+_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{3,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} - \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} V^{\text{CKM}}_{3,2} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} V^{\text{CKM}}_{3,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{3,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} V^{\text{CKM}}_{3,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{u}_1, b_2, W^+_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,3} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} V^{\text{CKM}}_{1,3} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} V^{\text{CKM}}_{1,3} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,3} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} V^{\text{CKM}}_{1,3} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{u}_1, d_2, W^+_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} V^{\text{CKM}}_{1,1} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} V^{\text{CKM}}_{1,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt{2} MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} V^{\text{CKM}}_{1,1} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{u}_1, s_2, W^+_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} - \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} V^{\text{CKM}}_{1,2} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} V^{\text{CKM}}_{1,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt{2} MZ^2 G_f \delta_{m_1, m_2} V^{\text{CKM}}_{1,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} V^{\text{CKM}}_{1,2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{b}_1, c_2, W^-_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{2,3^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{2,3^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{2,3^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt{2} MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{2,3^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{2,3^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{s}_1, t_2, W^-_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{3,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{3,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{3,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{3,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{3,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{s}_1, u_2, W^-_3$

$$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder } MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i \text{CFTn0 ChiralOrder } MZ \delta_{m_1, m_2} \sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} \\
& \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} MZ^2 G_f \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}} - \\
& \frac{2 i 2^{3/4} \pi \text{aEWM } \delta_{m_1, m_2} \sqrt{MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} + MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM}}
\end{aligned}$$

- $\bar{b}_1, b_2, A_3$

$$- \frac{i \sqrt[4]{2} \delta_{m_1, m_2} \sqrt{MZ^2 G_f} \sqrt{-MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} - MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{3 MZ^2 G_f}$$

- $\bar{d}_1, d_2, A_3$

$$- \frac{i \sqrt[4]{2} \delta_{m_1, m_2} \sqrt{MZ^2 G_f} \sqrt{-MZ \left(\sqrt{G_f (MZ^2 G_f - 2\sqrt{2}\pi \text{aEWM})} - MZ G_f \right) V^{\text{CKM}}_{1,2^*} \gamma^{\mu 3} . P_{-s_1, s_2}}}{3 MZ^2 G_f}$$

- $\bar{s}_1, s_2, A_3$

$$\begin{aligned}
& \frac{i \sqrt[4]{2} \sqrt{G_f MZ^2} \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} MZ}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{4 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{2 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi) MZ} + \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi) MZ}
\end{aligned}$$

- $\bar{s}_1, s_2, Z_3$

$$\begin{aligned}
& \frac{2 i \sqrt[4]{2} \text{CFTn0 ChiralOrder} G_f \sqrt{G_f MZ^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} MZ^2}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{2 i \sqrt[4]{2} G_f \sqrt{G_f MZ^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} MZ^2}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} G_f \sqrt{G_f MZ^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} MZ^2}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{i \sqrt[4]{2} G_f \sqrt{G_f MZ^2} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} MZ^2}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} MZ}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{i \sqrt[4]{2} \sqrt{G_f MZ^2} \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2} MZ}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} MZ}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{i \sqrt[4]{2} \sqrt{G_f MZ^2} \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2} MZ}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{4 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{4 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} - \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{2 i 2^{3/4} \text{aEWM CFTn0 ChiralOrder} \sqrt{G_f MZ^2} \pi \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} + \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{-s_1, s_2}}{3 G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi) MZ} + \\
& \frac{2 i 2^{3/4} \text{aEWM} \sqrt{G_f MZ^2} \pi \sqrt{G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi)} \delta_{m_1, m_2} \gamma^{\mu 3} . P_{+s_1, s_2}}{3 G_f (G_f MZ^2 - 2\sqrt{2} \text{aEWM } \pi) MZ}
\end{aligned}$$

- $\bar{\nu}_{e,1}, \nu_{e,2}, Z_3$

$$i \sqrt[4]{2} \sqrt{MZ^2} G_f \gamma^{\mu 3} . P_{-s_1, s_2} - i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{MZ^2} G_f \gamma^{\mu 3} . P_{-s_1, s_2}$$

- $$i \sqrt[4]{2} \sqrt{M Z^2 G_f \gamma^{\mu 3}} . P_{-s_1, s_2} - i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{M Z^2 G_f \gamma^{\mu 3}} . P_{-s_1, s_2}$$

- $$i \sqrt[4]{2} \sqrt{M Z^2 G_f \gamma^{\mu 3}} . P_{-s_1, s_2} - i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \sqrt{M Z^2 G_f \gamma^{\mu 3}} . P_{-s_1, s_2}$$

- $$\begin{aligned}
& - \frac{i \text{ CFTn0 ChiralOrder MZ}^2 G_f \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right)} \\
& + \frac{i \text{ CFTn0 ChiralOrder MZ} \sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right)} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right)} + \\
& + \frac{i 2^{3/4} \pi \text{ aEWM CF'Tn0 ChiralOrder} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM}} + \\
& - \frac{i \sqrt[4]{2} \text{MZ}^2 G_f \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM}} \\
& - \frac{2 i 2^{3/4} \pi \text{ aEWM} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{ aEWM}}
\end{aligned}$$

- $$\begin{aligned} & - \frac{i \text{CFTn0 ChiralOrder MZ}^2 G_f \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} - \\ & \frac{i \text{CFTn0 ChiralOrder MZ} \sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} + \\ & \frac{i 2^{3/4} \pi \text{aEWM CFTn0 ChiralOrder} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ & \frac{i \sqrt[4]{2} \text{MZ}^2 G_f \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ & \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} \end{aligned}$$

- $$\begin{aligned}
& - \frac{i \text{CFTn0 ChiralOrder} \text{MZ}^2 G_f \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3 . P_{-s_1, s_2}}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} - \\
& \frac{i \text{CFTn0 ChiralOrder} \text{MZ} \sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} \sqrt{\text{MZ} \left(\sqrt{G_f \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right) + \text{MZ} G_f} \right) \gamma^{\mu 3 . P_{-s_1, s_2}}}}{2^{3/4} \left(\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM} \right)} +
\end{aligned}$$

$$\frac{i \sqrt[4]{2} M Z^2 G_f \sqrt{M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + M Z G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} -$$

$$\frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + M Z G_f \right) \gamma^{\mu 3} . P_{-s_1, s_2}}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}}$$

- $\bar{e}_1, e_2, A_3$

$$\frac{i \sqrt[4]{2} \sqrt{M Z^2 G_f} \sqrt{-M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - M Z G_f \right) \sqrt{M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + M Z G_f \right) \gamma_{s_1, s_2}^{\mu 3}}}{M Z^2 G_f}$$

- $\bar{\mu}_1, \mu_2, A_3$

$$\frac{i \sqrt[4]{2} \sqrt{M Z^2 G_f} \sqrt{-M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - M Z G_f \right) \sqrt{M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + M Z G_f \right) \gamma_{s_1, s_2}^{\mu 3}}}{M Z^2 G_f}$$

- $\bar{\tau}_1, \tau_2, A_3$

$$\frac{i \sqrt[4]{2} \sqrt{M Z^2 G_f} \sqrt{-M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - M Z G_f \right) \sqrt{M Z \left(\sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} + M Z G_f \right) \gamma_{s_1, s_2}^{\mu 3}}}{M Z^2 G_f}$$

- $\bar{e}_1, e_2, Z_3$

$$\frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} -$$

$$\frac{i \sqrt[4]{2} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} +$$

$$\frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{M Z G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} -$$

$$\frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} M Z^2 G_f \sqrt{M Z^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} +$$

$$\frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} +$$

$$\frac{2 i 2^{3/4} \pi \text{aEWM} \text{CFTn0 ChiralOrder} \sqrt{M Z^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{i \sqrt[4]{2} M Z^2 G_f \sqrt{M Z^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} -$$

$$\frac{i \sqrt[4]{2} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{M Z^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} +$$

$$\frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{M Z G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}$$

- $\bar{\mu}_1, \mu_2, Z_3$

$$\frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} -$$

$$\frac{i \sqrt[4]{2} M Z \sqrt{M Z^2 G_f} \sqrt{G_f (M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{M Z^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} +$$

$$\begin{aligned}
& \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{\text{MZ} G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \text{MZ}^2 G_f \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM} \text{CFTn0 ChiralOrder} \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{i \sqrt[4]{2} \text{MZ}^2 G_f \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\
& \frac{i \sqrt[4]{2} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ} G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}
\end{aligned}$$

- $\bar{\tau}_1, \tau_2, Z_3$

$$\begin{aligned}
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\
& \frac{i \sqrt[4]{2} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{-s_1, s_2}}{\text{MZ} G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} - \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \text{MZ}^2 G_f \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{i \sqrt[4]{2} \text{CFTn0 ChiralOrder} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM} \text{CFTn0 ChiralOrder} \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{i \sqrt[4]{2} \text{MZ}^2 G_f \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\
& \frac{i \sqrt[4]{2} \text{MZ} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\
& \frac{2 i 2^{3/4} \pi \text{aEWM} \sqrt{\text{MZ}^2 G_f} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})} \gamma^{\mu 3} . P_{+s_1, s_2}}{\text{MZ} G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}
\end{aligned}$$

- $\bar{c}_1, c_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu 3}$$

- $\bar{t}_1, t_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu 3}$$

- $\bar{u}_1, u_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu 3}$$

- $\bar{b}_1, b_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu 3}$$

- $\bar{d}_1, d_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu_3}$$

- $\bar{s}_1, s_2, g_3$

$$2 i \sqrt{\pi} \sqrt{\alpha_s} T_{m_1, m_2}^{a_3} \gamma_{s_1, s_2}^{\mu_3}$$

4-point vertices

- H_1, H_2, H_3, H_4

$$-3 i \sqrt{2} \text{Kappa4H} \text{MH}^2 G_f$$

- g_1, g_2, H_3, H_4

$$2 i \sqrt{2} \text{CGHNLO} \text{n2 ChiralOrder} p_1 \cdot p_2 G_f \delta_{a_1, a_2} \eta_{\mu_1, \mu_2} - \\ 2 i \sqrt{2} \text{CGHNLO} \text{n2 ChiralOrder} G_f \delta_{a_1, a_2} p_1^{\mu_2} p_2^{\mu_1}$$

- g_1, g_2, g_3, g_4

$$4 i \pi \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} + \\ 4 i \pi \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} + 4 i \pi \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\ 4 i \pi \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} - 4 i \pi \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\ 4 i \pi \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} - 8 i \alpha_s^2 \theta_s \epsilon_{\mu_1, \mu_2, \mu_3, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} + \\ 8 i \alpha_s^2 \theta_s \epsilon_{\mu_1, \mu_2, \mu_3, \mu_4} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} - 8 i \alpha_s^2 \theta_s \epsilon_{\mu_1, \mu_2, \mu_3, \mu_4} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}}$$

- g_1, g_2, g_3, H_4

$$2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} p_1^{\mu_3} - \\ 2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} p_2^{\mu_3} - \\ 2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} p_1^{\mu_2} + \\ 2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} p_3^{\mu_2} + \\ 2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} p_2^{\mu_1} - \\ 2 \sqrt{2} \sqrt{\pi} \text{CGHNLO} \text{n1 ChiralOrder} \sqrt{G_f} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} p_3^{\mu_1}$$

- H_1, H_2, W^+_3, W^-_4

$$\frac{8 i \pi \text{aEWM} \text{CFTn0 ChiralOrder} \text{Kappa2V} \text{MZ}^2 G_f \eta_{\mu_3, \mu_4}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ \frac{4 i \pi \text{aEWM} \text{CFTn0 ChiralOrder} \text{Kappa2V} \text{MZ} \eta_{\mu_3, \mu_4} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{2 i \sqrt{2} \text{CFTn0 ChiralOrder} \text{Kappa2V} \text{MZ}^4 G_f^2 \eta_{\mu_3, \mu_4}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{2 i \sqrt{2} \text{CFTn0 ChiralOrder} \text{Kappa2V} \text{MZ}^3 G_f \eta_{\mu_3, \mu_4} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{4 i \pi \text{aEWM} \text{Kappa2V} \text{MZ}^2 G_f \eta_{\mu_3, \mu_4}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \frac{4 i \pi \text{aEWM} \text{Kappa2V} \text{MZ} \eta_{\mu_3, \mu_4} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ \frac{i \sqrt{2} \text{Kappa2V} \text{MZ}^4 G_f^2 \eta_{\mu_3, \mu_4}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{i \sqrt{2} \text{Kappa2V} \text{MZ}^3 G_f \eta_{\mu_3, \mu_4} \sqrt{G_f (\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{\text{MZ}^2 G_f - 2 \sqrt{2} \pi \text{aEWM}}$$

- A_1, A_2, W^+_3, W^-_4

$$4 i \pi \text{aEWM} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} + 4 i \pi \text{aEWM} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} - 8 i \pi \text{aEWM} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4}$$

- $W^+_1, W^+_2, W^-_3, W^-_4$

$$\begin{aligned} & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \frac{2 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \frac{2 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{8 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \frac{4 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^4}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{2 i \sqrt{2} G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{2 i \sqrt{2} G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{8 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{4 i \sqrt{2} G_f \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^3}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \frac{8 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \frac{8 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \frac{16 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{32 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}^2}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{8 i \text{aEWM} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{8 i \text{aEWM CFTn0 ChiralOrder} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{8 i \text{aEWM} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{8 i \text{aEWM CFTn0 ChiralOrder} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} - \\ & \frac{16 i \text{aEWM} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} + \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder} \pi \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} \text{MZ}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi} \end{aligned}$$

- A_1, W^+_2, W^-_3, Z_4

$$\frac{2 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{-\text{MZ} \left(\sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} - G_f \text{MZ} \right)} \sqrt{\text{MZ} \left(G_f \text{MZ} + \sqrt{G_f (G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi)} \right)}}{G_f \text{MZ}^2 - 2 \sqrt{2} \text{aEWM} \pi}$$

[illegible]

$$\frac{4 i \text{aEWM } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \sqrt{-MZ \left(\sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} - G_f MZ \right)} \sqrt{MZ \left(G_f MZ + \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \right)}}{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi) MZ}$$

- H_1, H_2, Z_3, Z_4

$$-4 i \sqrt{2} \text{CFTn0 ChiralOrder Kappa2V } MZ^2 G_f \eta_{\mu_3, \mu_4} + 4 i \sqrt{2} \text{CFTn2 ChiralOrder } MZ^2 G_f \eta_{\mu_3, \mu_4} + 2 i \sqrt{2} \text{Kappa2V } MZ^2 G_f \eta_{\mu_3, \mu_4}$$

- $W^+_{-1}, W^-_{-2}, Z_3, Z_4$

$$\begin{aligned} & - \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \frac{2 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \frac{2 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{8 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f^2 \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \frac{4 i \sqrt{2} G_f^2 \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^4}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{2 i \sqrt{2} G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{4 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{2 i \sqrt{2} G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{8 i \sqrt{2} \text{CFTn0 ChiralOrder } G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{4 i \sqrt{2} G_f \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^3}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \frac{12 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \frac{12 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \frac{24 i \text{aEWM } G_f \pi \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{32 i \text{aEWM CFTn0 ChiralOrder } G_f \pi \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ^2}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{8 i \text{aEWM } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{8 i \text{aEWM CFTn0 ChiralOrder } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{8 i \text{aEWM } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{8 i \text{aEWM CFTn0 ChiralOrder } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{16 i \text{aEWM } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \\ & \frac{16 i \text{aEWM CFTn0 ChiralOrder } \pi \sqrt{G_f (G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi)} \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} MZ}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \\ & \frac{8 i \sqrt{2} \text{aEWM}^2 \pi^2 \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3}}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} + \frac{8 i \sqrt{2} \text{aEWM}^2 \pi^2 \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4}}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} - \frac{16 i \sqrt{2} \text{aEWM}^2 \pi^2 \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4}}{G_f MZ^2 - 2 \sqrt{2} \text{aEWM } \pi} \end{aligned}$$

5-point vertices

- g_1, g_2, H_3, H_4, H_5

$$6 i 2^{3/4} \text{CGHNLOn3 ChiralOrder } p_1 \cdot p_2 G_f^{3/2} \delta_{a_1, a_2} \eta_{\mu_1, \mu_2} - \\ 6 i 2^{3/4} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \delta_{a_1, a_2} p_1^{\mu_2} p_2^{\mu_1}$$

- g_1, g_2, g_3, H_4, H_5

$$4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} p_1^{\mu_3} - \\ 4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} p_2^{\mu_3} - \\ 4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} p_1^{\mu_2} + \\ 4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} p_3^{\mu_2} + \\ 4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} p_2^{\mu_1} - \\ 4 \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} p_3^{\mu_1}$$

- g_1, g_2, g_3, g_4, H_5

$$4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} + \\ 4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} + \\ 4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\ 4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} - \\ 4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\ 4 i \sqrt[4]{2} \pi \text{CGHNLOn1 ChiralOrder } \sqrt{G_f} \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}}$$

- $H_1, H_2, H_3, W^+_4, W^-_5$

$$\frac{24 i \sqrt[4]{2} \pi \text{aEWM CFCn3 CFTn0 ChiralOrder } MZ^2 G_f^{3/2} \eta_{\mu_4, \mu_5}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ \frac{12 i \sqrt[4]{2} \pi \text{aEWM CFCn3 CFTn0 ChiralOrder } MZ \sqrt{G_f} \eta_{\mu_4, \mu_5} \sqrt{G_f (MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{6 i 2^{3/4} \text{CFCn3 CFTn0 ChiralOrder } MZ^4 G_f^{5/2} \eta_{\mu_4, \mu_5}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{6 i 2^{3/4} \text{CFCn3 CFTn0 ChiralOrder } MZ^3 G_f^{3/2} \eta_{\mu_4, \mu_5} \sqrt{G_f (MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{12 i \sqrt[4]{2} \pi \text{aEWM CFCn3 } MZ^2 G_f^{3/2} \eta_{\mu_4, \mu_5}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} - \\ \frac{12 i \sqrt[4]{2} \pi \text{aEWM CFCn3 } MZ \sqrt{G_f} \eta_{\mu_4, \mu_5} \sqrt{G_f (MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \frac{3 i 2^{3/4} \text{CFCn3 } MZ^4 G_f^{5/2} \eta_{\mu_4, \mu_5}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}} + \\ \frac{3 i 2^{3/4} \text{CFCn3 } MZ^3 G_f^{3/2} \eta_{\mu_4, \mu_5} \sqrt{G_f (MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM})}}{MZ^2 G_f - 2 \sqrt{2} \pi \text{aEWM}}$$

- H_1, H_2, H_3, Z_4, Z_5

$$-12 i 2^{3/4} \text{CFCn3 CFTn0 ChiralOrder } MZ^2 G_f^{3/2} \eta_{\mu_4, \mu_5} + \\ 6 i 2^{3/4} \text{CFCn3 } MZ^2 G_f^{3/2} \eta_{\mu_4, \mu_5} + 12 i 2^{3/4} \text{CFTn3 ChiralOrder } MZ^2 G_f^{3/2} \eta_{\mu_4, \mu_5}$$

6-point vertices

- $g_1, g_2, g_3, H_4, H_5, H_6$

$$\begin{aligned}
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} P_1^{\mu_3} - \\
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_2} P_2^{\mu_3} - \\
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} P_1^{\mu_2} + \\
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_1, \mu_3} P_3^{\mu_2} + \\
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} P_2^{\mu_1} - \\
& 12 \, 2^{3/4} \sqrt{\pi} \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \sqrt{\alpha_s} f_{a_1, a_2, a_3} \eta_{\mu_2, \mu_3} P_3^{\mu_1}
\end{aligned}$$

- $g_1, g_2, g_3, g_4, H_5, H_6$

$$\begin{aligned}
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} + \\
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} + \\
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} - \\
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\
& 8 \, i \sqrt{2} \pi \text{CGHNLOn2 ChiralOrder } G_f \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}}
\end{aligned}$$

7-point vertices

- $g_1, g_2, g_3, g_4, H_5, H_6, H_7$

$$\begin{aligned}
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}} + \\
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_4} \eta_{\mu_2, \mu_3} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} + \\
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_3} \eta_{\mu_2, \mu_4} f_{a_1, a_2, \text{Gluon1}} f_{a_3, a_4, \text{Gluon1}} - \\
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_4, \text{Gluon1}} f_{a_2, a_3, \text{Gluon1}} - \\
& 24 \, i \, 2^{3/4} \pi \text{CGHNLOn3 ChiralOrder } G_f^{3/2} \alpha_s \eta_{\mu_1, \mu_2} \eta_{\mu_3, \mu_4} f_{a_1, a_3, \text{Gluon1}} f_{a_2, a_4, \text{Gluon1}}
\end{aligned}$$